Celestial Mechanics

01.01 · Original practice problems

Show your reasoning, use SI units where appropriate, and state any assumptions. Constants are supplied in the questions that need them.

CONTENTS

01 Circular orbit and escape

02 An elliptical journey

03 A binary and its barycentre

04 Tides and equilibrium points

A MAIS Physics & Astronomy Club study resource. These are original exercises, not official IOAA problems.

PROBLEMS

01 Circular orbit and escape

A planet has mass 6.0×10^{24} kg and radius 6.4×10^{6} m. A small satellite moves in a circular orbit 4.0×10^{5} m above the surface. Use $G = 6.67 \times 10^{-11}$ N m² kg⁻². (a) Find its orbital speed and period. (b) Find the escape speed at the orbit. (c) Explain why the satellite is not weightless in the sense of experiencing no gravity.

02 An elliptical journey

A comet orbits a star of mass 2.0×10^{30} kg. Its periapsis and apoapsis distances are 1.0×10^{11} m and 5.0×10^{11} m. Use $G = 6.67 \times 10^{-11}$ N m² kg⁻². (a) Find the semimajor axis and eccentricity. (b) Find the orbital period. (c) Find the ratio of the speeds at periapsis and apoapsis, explaining the conservation law used.

03 A binary and its barycentre

Two stars of masses 2.0 and 1.0 solar masses move on circular orbits about their common centre of mass. Their separation is 3.0 AU. Take 1 AU = 1.50×10^{11} m, one solar mass = 2.0×10^{30} kg and $G = 6.67 \times 10^{-11}$ N m² kg⁻². (a) Find the distance of each star from the barycentre. (b) Find the orbital period. (c) Which star has the larger orbital speed, and by what factor?

04 Tides and equilibrium points

A fluid moon of density ρ orbits a planet of radius R and density 8ρ. For this problem, use the approximate fluid Roche limit $d = 2.44\,R(\rho_{planet}/\rho_{moon})^{1/3}$. (a) Calculate d/R. (b) Explain physically why tidal disruption can occur inside this distance. (c) Describe where the L1 point lies in a planet–moon system and explain why it is an equilibrium point in the rotating frame. Does equilibrium necessarily imply stability?